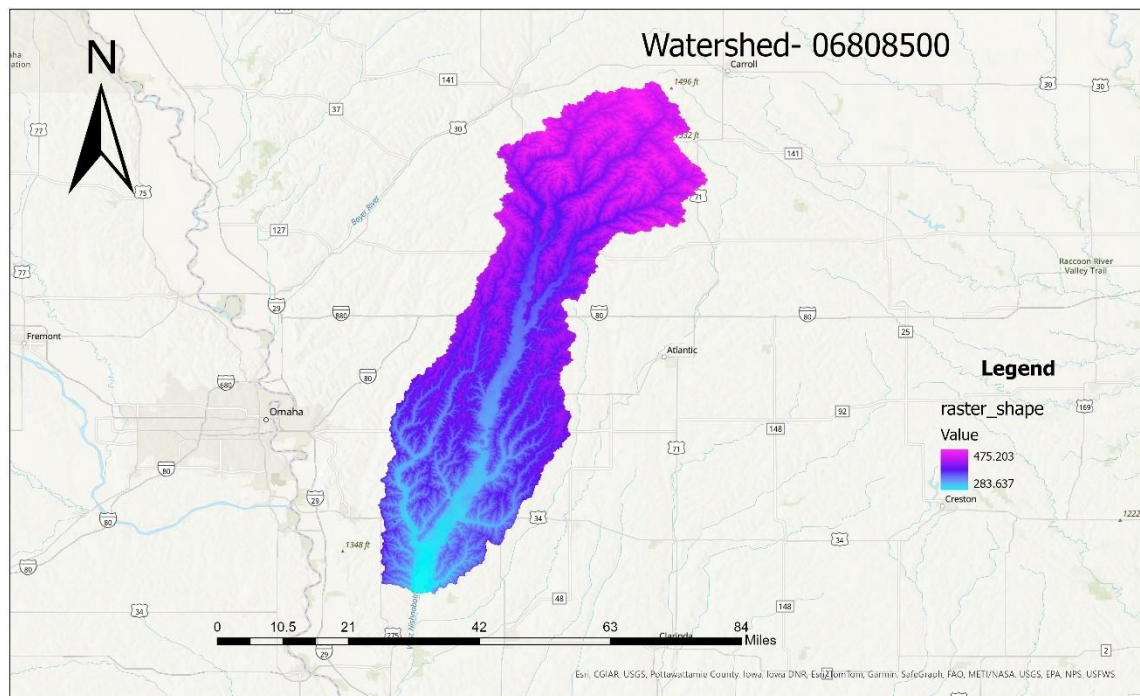


# Hydraulic Modelling of the West Nishnabonta River at Randolph, Iowa



By: Maaz Bin Naseer

## Abstract:

This study aimed to calibrate and validate a hydrological model of the West Nishnabotna River Watershed in Randolph, Iowa, using HEC-HMS. The objective was to improve the accuracy of simulated hydrographs for a 100-year storm event. The findings emphasized the importance of parameter refinement in predictive modeling and demonstrated the utility of hydrological models for flood risk assessment and watershed management.

## Introduction:

This project focuses on hydrologic modeling of the **West Nishnabotna River Watershed** at Randolph, Iowa, to understand watershed dynamics and manage flood risks. Using the Hydrologic Modeling System (HEC-HMS), the study aims to calibrate the model with historical rainfall and runoff data and simulate the watershed's response to a 100-year design storm. The workflow involved delineating the watershed and sub-basins using topographic data, integrating land use and soil properties to characterize surface and subsurface behavior, and driving the model with historical rainfall data. Calibration was performed to align simulated runoff with observed streamflow, ensuring accuracy. The model was further validated using a similar rainfall event, confirming its reliability.

The calibrated HEC-HMS model was then applied to simulate the hydrologic response to a 100-year storm, providing critical insights into peak discharge and flood risks. These discharge values were subsequently used in HEC-RAS to model flood inundation, helping identify flood-prone areas and informing land use planning, emergency response, and infrastructure design. This integrated approach underscores the critical role of accurate calibration in predictive modeling, offering essential tools for effective flood risk management and informed decision-making within the West Nishnabotna River Watershed.

## Study Area:

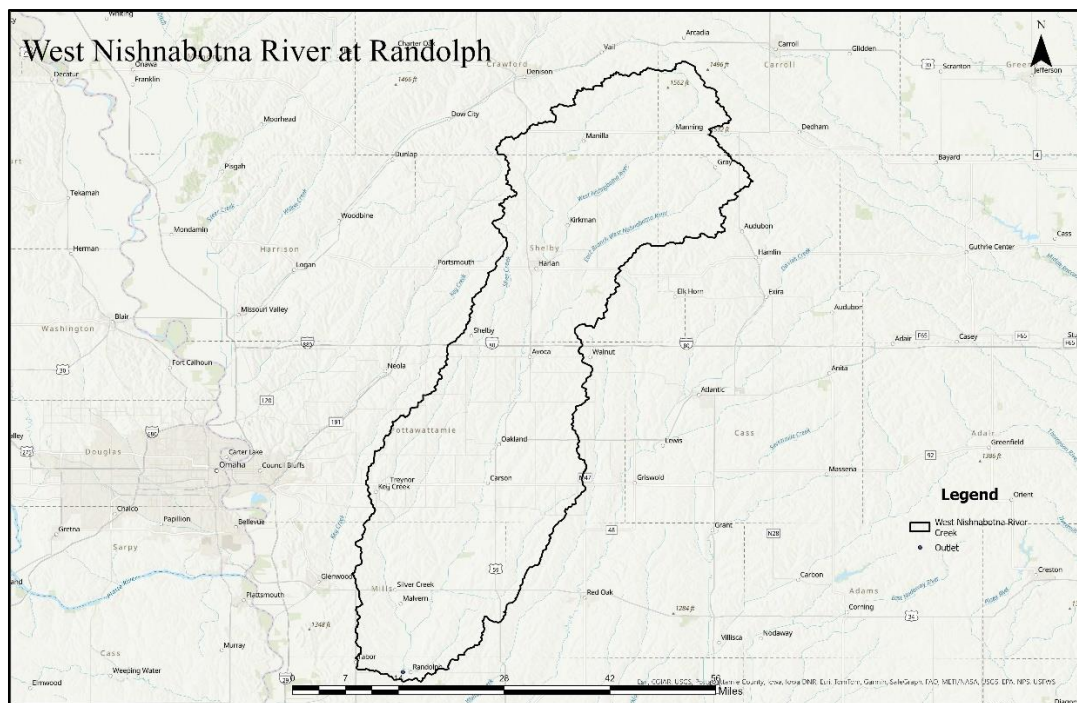


Figure 1 West Nishnabotna River Watershed

The study area is the **West Nishnabotna River Watershed**, which encompasses approximately 3,447.42 square kilometers and is part of the larger Nishnabotna River Watershed within the Missouri River Basin. This watershed

## CE 549: Computational Watershed Hydrology

is in southwestern Iowa and northwestern Missouri, with its primary drainage point at the USGS monitoring station 06808500, situated on the West Nishnabotna River at Randolph, Iowa. The West Nishnabotna River is currently 80% cropland and is used heavily for recreation and tourism. (Iowa Watershed Approach, 2024)

The West Nishnabotna River originates in southwestern Iowa and flows southward, contributing to the Nishnabotna River before ultimately joining the Missouri River. It plays a critical role in water resource management, providing water for agriculture and supporting local ecosystems. A map of the study area, including the watershed boundary and its significant hydrological features, is presented in Figure 1, illustrating the spatial extent and connectivity of the watershed to the broader Missouri River system with the watershed gauge located at 40.8732778 Latitude and -95.581444 Longitude.

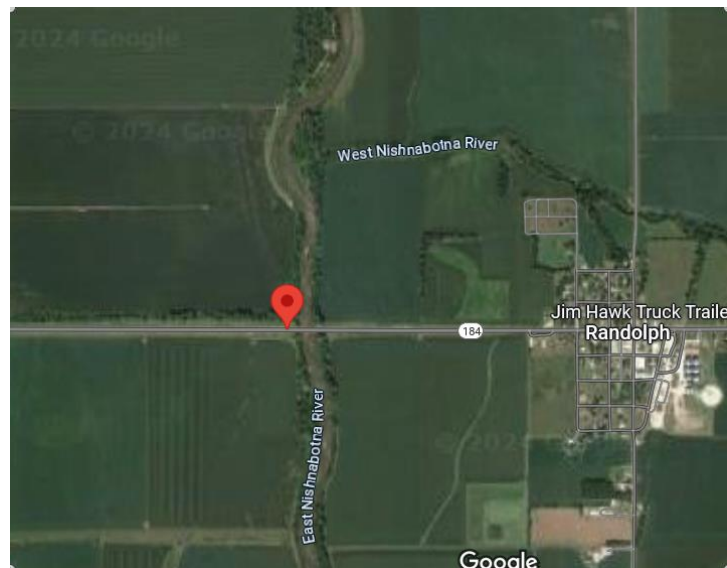


Figure 2 Location of the Water Gauge

Along the river, while annual row crops border certain areas, much of the riparian buffer is composed of perennial vegetation, including woodlands, grasslands, and wetlands. Silver maple is the dominant species in many areas near the water, especially where the riverbanks are less steep. Willows thrive on the inside bends of the river, creating a “willow wedge” with larger, potentially older willows farther from the water. Cottonwoods are common and, on occasion, sycamores can also be found. On the benches above the steeper slopes, trees such as walnuts, hackberries, and boxelders are present. The understory often includes water hemlock and elderberries, while open areas feature reed canary grass and brome grass. Wet areas are frequently dominated by solid stands of stinging nettles, with giant ragweed filling in behind them. Vines such as grapevine and occasional moonseed or green briar can be seen climbing snags and hanging over the riverbanks. Boxelder and mulberries are also common along the sloped banks, with walnuts occasionally appearing on the benches above the river in some areas. (Weebly, 2024)

## Data:

The data that were used to make the DEM came from the USGS National Elevation Dataset (USGS National Elevation, 2024). The shapefile was generated from the website called cyberfaces (Cyberfaces, 2024). Land use and soil data was taken from the National Land Cover Database (USGS National Land Cover, 2024). The stream flow data was taken from USGS Current Water Data for the Nation (USGS, 2024) and the precipitation flow data was taken from Hydrometeorological Design Studies center (NOAA, 2024). The streamflow and DEM were used to create a stream network and watershed delineation. Initially the original DEM file came in geographic coordinate system, which then later got converted to Projected Co-ordinate system NAD 1983 UTM Zone 15N. The spatial resolution of the cell was 30m x 30m for the raster.

## Methodology:

The project employed an integrated workflow leveraging cloud-based Jupyter Notebooks and desktop GIS tools to conduct hydrologic and hydraulic modeling for the West Nishnabotna River Watershed. The watershed polygon served as the foundational layer to retrieve the Digital Elevation Model (DEM) and Land Use Land Cover (LULC) datasets, capturing the area's topography and land cover characteristics. LULC data, combined with gSSURGO soil information, was analyzed in ArcGIS to calculate Curve Number (CN) values, which represent the watershed's runoff potential.

The DEM was imported into HEC-HMS for stream delineation, sub-basin identification, and parameterization, including CN values and lag times, essential for simulating hydrologic responses to precipitation. The calibrated model was used to simulate a 100-year storm event using precipitation data from NOAA's Precipitation Frequency Data Server, providing peak discharge values for hydraulic modeling. HEC-RAS utilized these peak discharges, along with cross-sectional data, to simulate river hydraulics and floodplain inundation. The outputs, including water surface elevations and inundation boundaries, offered critical insights into flood risks and water management strategies for the West Nishnabotna River Watershed.

### 4.1. HEC-HMS and calibration

In this section, HEC-HMS model; terrain data was used to load the DEM for water delineation, with a stream threshold of 4% of the total stream network threshold. In my project area, it created 13 subbasins, with each subbasin containing curve number, lag times, and routing co-efficient. The initial CN values were selected from the ARCGIS during the watershed creation, initial Muskingum routing was selected to be K (0.5) and X (0.25), initial calculation of the lag time using the equation:

$$T_c = \frac{l^{0.8}(S + 1)^{0.7}}{1140Y^{0.5}}$$

Where:

$T_c$  = time of concentration in hours,  $l$  = flow length of the longest flow path in ft,  $Y$  = average watershed land slope, %  $S$  = maximum potential retention in inches =  $(1000/CN) - 10$ ,  $CN$  = SCS Curve Number ( $0 < CN < 99$ )

The rainfall event selected for this study occurred in February 2020 and was characterized by a heavy, one-day rainfall event. This event was chosen because it provided consistent and complete data compared to other events within the recent five-year period. Upon reviewing precipitation records from the past five years, it was determined that many events contained data gaps or inconsistencies that could compromise the reliability of model calibration. However, the February 2020 event stood out due to its consistency, completeness, and significant hydrologic impact, making it an ideal candidate for accurately calibrating the model to simulate watershed response. Furthermore, this study will validate the February 2020 rainfall event by examining consistent precipitation and runoff data from the past 10 years and the dates that were chosen for validation were from 29 March 2017 to 1<sup>st</sup> April 2017. By analyzing historical records, the validation process will determine if the February 2020 event accurately reflects the watershed's long-term hydrologic behavior. This approach not only reinforces the credibility of the calibrated model but also provides a stronger foundation for predicting and managing extreme hydrologic events in the West Nishnabotna River Watershed.

The final step is to perform a 100-year storm analysis, which is critical for understanding the watershed's response to extreme weather conditions. This analysis provides key insights into peak discharge, flood risks, and potential inundation areas, enabling the design of effective flood management strategies. Additionally, it helps inform infrastructure development, emergency response planning, and long-term watershed management to protect communities and ecosystems from catastrophic flooding events.

Manual calibration was found to be more effective and easier to perform compared to automatic calibration, resulting in nearly a 80% improvement in curve time and lag time. The calibration process involved systematically



## CE 549: Computational Watershed Hydrology

adjusting key hydrological and routing parameters to achieve a better fit between the observed and simulated flow hydrographs. The curve number (CN) was modified to reflect the watershed's runoff potential, where it was decreased for areas with higher infiltration potential or increased for more impervious regions. This adjustment helped simulate runoff volumes more accurately. Lag time, representing the time it takes for runoff to travel from the watershed centroid to the outlet, was increased to better match the timing of observed peak flows. In the reach routing process, the Muskingum X parameter, which defines the shape of the storage-discharge relationship, was adjusted to reduce peak flow magnitudes and improve attenuation. Additionally, Muskingum K, representing the travel time of water through river reaches, was increased to slow the flow and align the recession limb with the observed hydrograph. These iterative adjustments provided flexibility to visually inspect and refine the model's performance, resulting in a calibrated simulation that closely aligned with the observed hydrograph in terms of peak flow magnitude, timing, and the recession curve.

|             | CN value (unitless) | Lag Time (minutes) |
|-------------|---------------------|--------------------|
| Element     | Uncalibrated        | Uncalibrated       |
| Subbasin-1  | 85.36               | 446.6496           |
| Subbasin-10 | 80.45               | 249.9552           |
| Subbasin-11 | 86.95               | 307.7088           |
| Subbasin-12 | 75.58               | 270.576            |
| Subbasin-13 | 82.65               | 301.6128           |
| Subbasin-2  | 85.56               | 492.8832           |
| Subbasin-3  | 87.96               | 586.8288           |
| Subbasin-4  | 83.26               | 460.2624           |
| Subbasin-5  | 81.21               | 425.8848           |
| Subbasin-6  | 80.15               | 421.8144           |
| Subbasin-7  | 79.96               | 457.488            |
| Subbasin-8  | 89.36               | 309.3504           |
| Subbasin-9  | 84.47               | 372.3936           |

Table 1 Initial Value for Curve and Lag Time

## 4.2. HEC-RAS:

For the HEC-RAS model, a 7 km stretch centered around the outlet was selected, covering both upstream and downstream areas to focus the analysis on the steady flow transition along the river. The **Raster Mapper** tool in HEC-RAS was used to define key hydraulic features, including the banks, flow paths, stream centerline, and cross sections for the West Nishnabotna River. Additionally, the bridge near the West Nishnabotna River gauge was modeled with two piers, and adjustments to the bridge data table were made. At the upstream boundary, the peak discharge derived from the HEC-HMS model was used to simulate flow entering the river at its head. For the downstream boundary, a normal depth condition with a slope of 0.002 was applied to reflect flow behavior at the outlet. Manning's roughness coefficients were carefully assigned, with a value of 0.035 for the main channel and 0.06 for the floodplain, based on land use and riverbed characteristics. The steady-flow simulation incorporated the 100-year peak discharge values obtained from HEC-HMS, and the final inundation area was derived from the simulation results.

## Results and Discussion:

### 4.1. HEC-HMS and calibration results

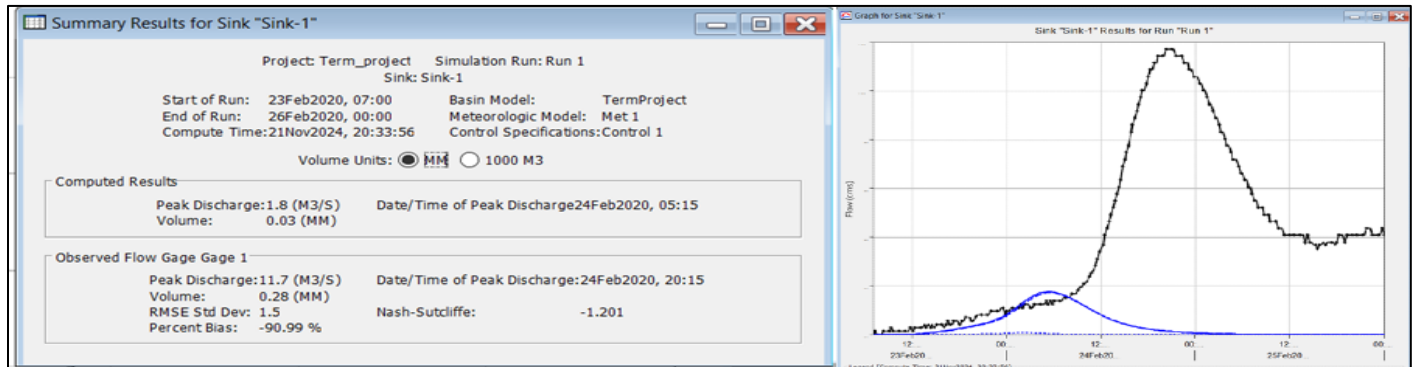


Figure 3 Uncalibrated event for 2020 Feb event

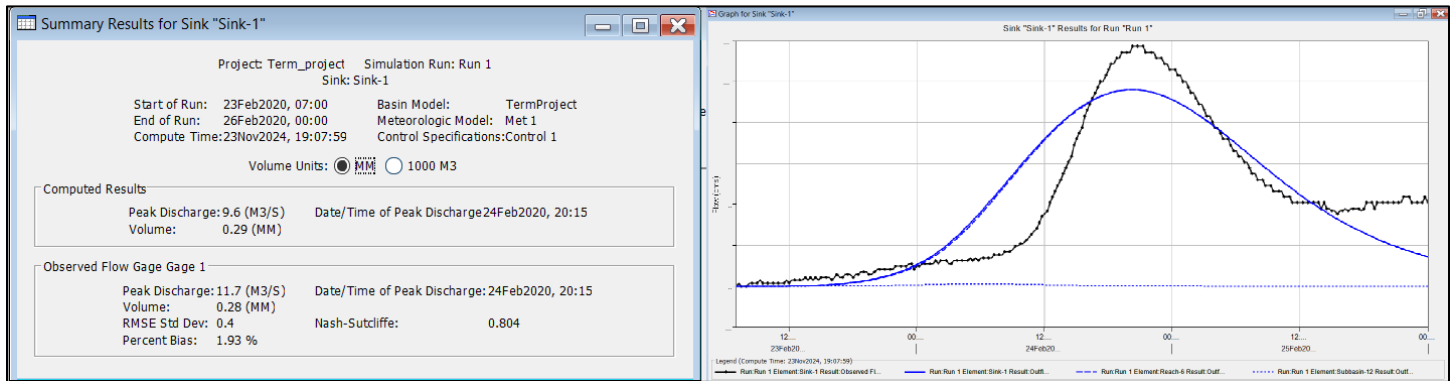


Figure 4 Calibrated for Feb 2020 event

From the results shown in the Figure 3, the simulated peak discharge for the sink is **1.8 m<sup>3</sup>/s**, occurring at **05:15 on February 24, 2020**, with a total volume of **0.03 million m<sup>3</sup>**. In comparison, the observed flow gauge data reports a significantly higher peak discharge of **11.17 m<sup>3</sup>/s**, occurring later at **20:15 on the same day**, with a total volume of **0.28 million m<sup>3</sup>**. The Nash-Sutcliffe efficiency for this simulation is **-1.201**, indicating a poor fit between the observed and simulated hydrographs. The RMSE (Root Mean Square Error) is **1.5**, suggesting notable discrepancies in flow predictions, and the percent bias is **-90.99%**, signifying an underestimation of the discharge values. In the Figure 4, the simulated peak discharge improves significantly to **9.46 m<sup>3</sup>/s**, now occurring at **20:15 on February 24, 2020**, closely matching the observed peak discharge of **11.17 m<sup>3</sup>/s** at the same time. The total simulated volume increases to **0.29 million m<sup>3</sup>**, aligning more closely with the observed volume of **0.28 million m<sup>3</sup>**. The Nash-Sutcliffe efficiency improves dramatically to **0.804**, indicating a much better fit between the observed and simulated hydrographs. Additionally, the RMSE is reduced to **0.42**, and the percent bias drops to **1.93%**, showcasing a more accurate simulation compared to the observed data. This suggests that the adjustments made in the second simulation better capture the hydrological behavior of the watershed.

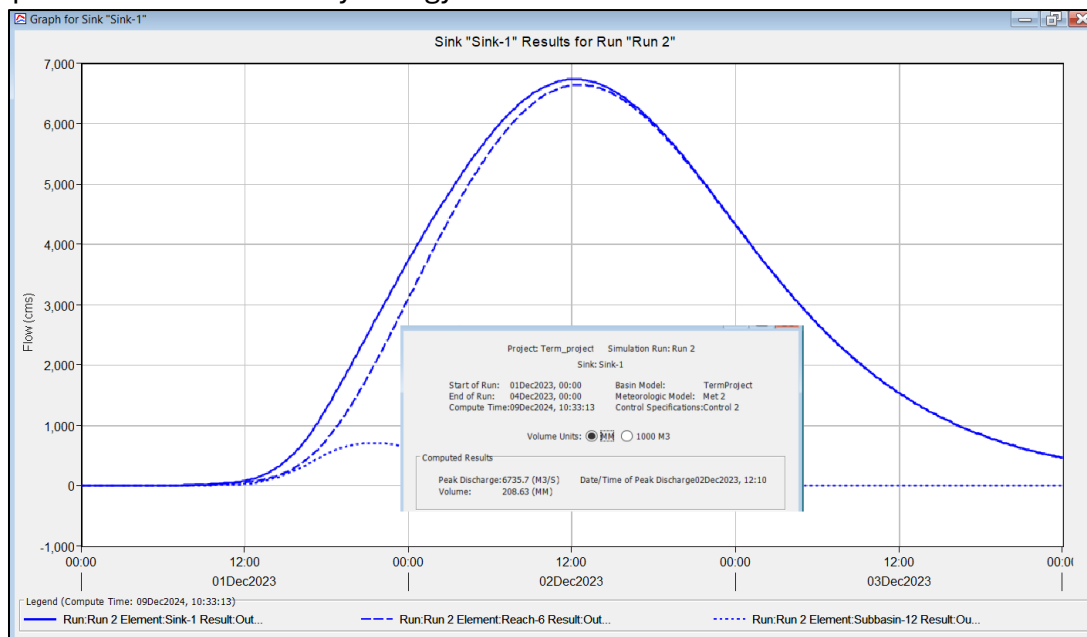


Figure 5 100-year storm analysis hydrograph

The graph and summary results represent the 100-year peak discharge simulation for Sink-1 under **Run 2**. The hydrograph illustrates the flow behavior over the simulation period, starting from **00:00 on December 1, 2023**, and ending on **December 4, 2023**. The peak discharge of **6735.7 m<sup>3</sup>/s** is observed at **12:10 on December 2, 2023**, indicating the most significant flow event during the simulation.

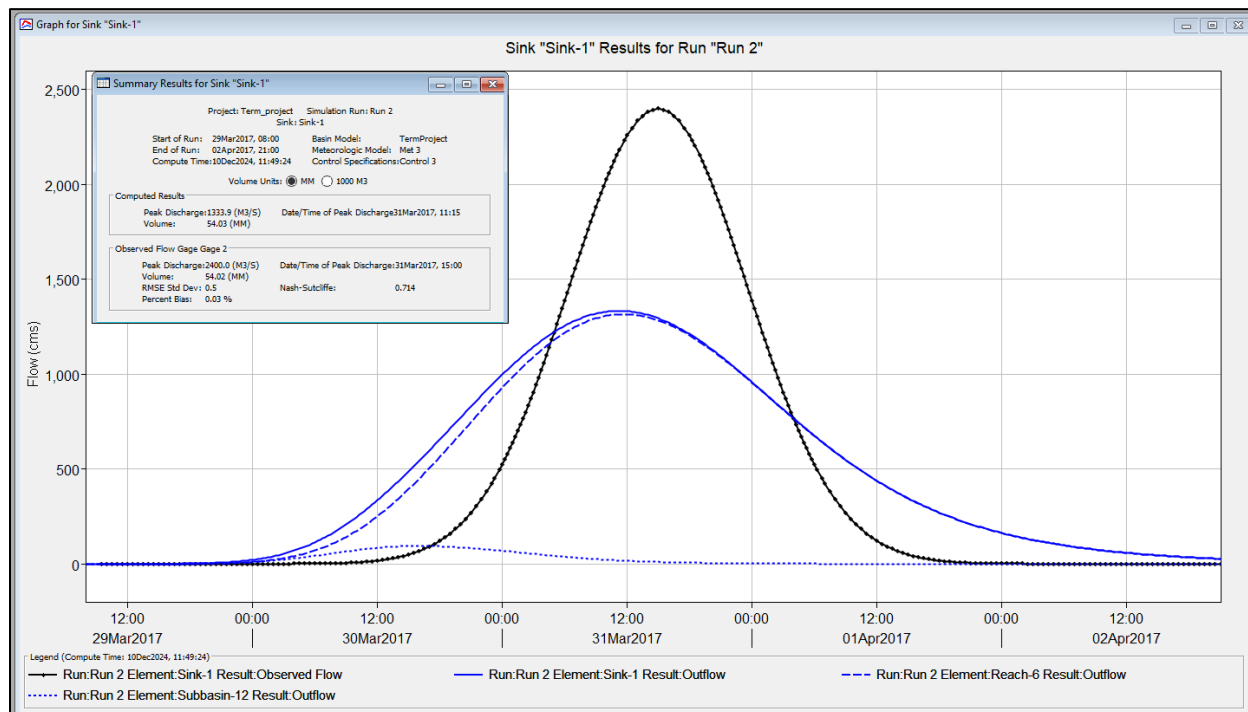


Figure 6 Calibrated and Validated Model

The figure illustrates the observed and simulated hydrographs for "Run 2" in the HEC-HMS model. The black curve represents the observed flow, while the blue curves indicate the simulated flow components, including the sink outflow and reach outflow. The observed peak discharge was recorded on **31 March 2017 at 11:55**, with a

## CE 549: Computational Watershed Hydrology

value of **2,401.94 m<sup>3</sup>/s**, as shown in the summary results. The simulated hydrograph aligns closely with the observed flow, particularly in the rising limb and peak timing, demonstrating accurate calibration. The model performance metrics, such as a **Percent Bias (PBIAS)** of **0.03%** and a **Nash-Sutcliffe Efficiency (NSE)** of **0.744**, indicate that the model has an excellent fit with negligible systematic error. The very low PBIAS value highlights the model's ability to effectively replicate both the flow magnitude and timing. This suggests that the adjustments to parameters like curve number, lag time, and Muskingum routing were effective in achieving high model accuracy

|             | Curve Number<br>(Unitless) | Lag Time<br>(mins) |         | Muskingum K | Muskingum X |
|-------------|----------------------------|--------------------|---------|-------------|-------------|
| Element     | Calibrated                 | Calibrated         | Element | Calibrated  | Calibrated  |
| Subbasin-1  | 99                         | 893.2992           | Reach-1 | 3           | 0.1         |
| Subbasin-10 | 98                         | 499.9104           | Reach-2 | 3           | 0.1         |
| Subbasin-11 | 98                         | 615.4176           | Reach-3 | 3           | 0.1         |
| Subbasin-12 | 88                         | 541.152            | Reach-4 | 3           | 0.1         |
| Subbasin-13 | 99                         | 603.2256           | Reach-5 | 3           | 0.1         |
| Subbasin-2  | 99                         | 985.7664           | Reach-6 | 3           | 0.1         |
| Subbasin-3  | 99                         | 1173.6576          |         |             |             |
| Subbasin-4  | 99                         | 920.5248           |         |             |             |
| Subbasin-5  | 99                         | 851.7696           |         |             |             |
| Subbasin-6  | 99                         | 843.6288           |         |             |             |
| Subbasin-7  | 99                         | 914.976            |         |             |             |
| Subbasin-8  | 98                         | 618.7008           |         |             |             |
| Subbasin-9  | 99                         | 744.7872           |         |             |             |

Table 2 Calibrated Values

## 4.2. HEC-RAS results:

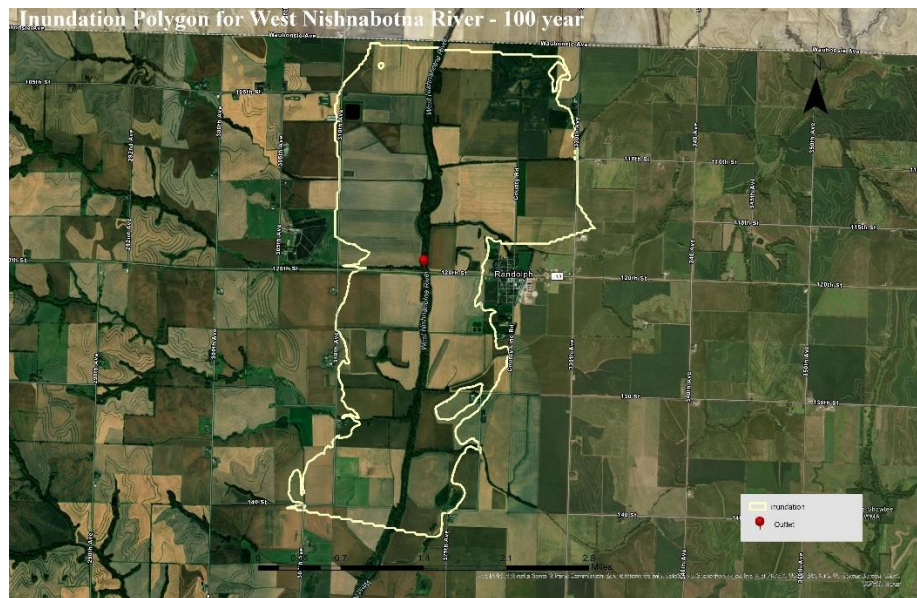


Figure 7 Inundation Area Polygon

The inundation polygon highlights that floodwaters primarily affect agricultural land, posing significant economic risks such as crop losses, soil degradation, and disruptions to farming operations, including potential damage to irrigation systems and equipment. The city of Randolph, IA, located near the inundation zone, may not be entirely within the flooded area but remain at risk of indirect impacts, including access road blockages, utility disruptions,



## CE 549: Computational Watershed Hydrology

and possible inundation of low-lying sections. These issues could isolate residents and complicate emergency response efforts.

Additionally, a critical bridge near the inundation area faces risks of structural damage due to prolonged flooding, potentially disrupting key transportation routes for residents and agricultural activities. Upstream of the infrastructure in the area, the Bass Landing Field is also a notable infrastructure at risk. If the airfield becomes inundated, it could impede its operations and limit its utility for emergencies or general use. To mitigate these risks, it is crucial to conduct structural and operational assessments of the bridge and airfield, plan alternate routes for transportation, and implement a localized flood management strategy to safeguard Randolph, the agricultural land, and critical infrastructure.

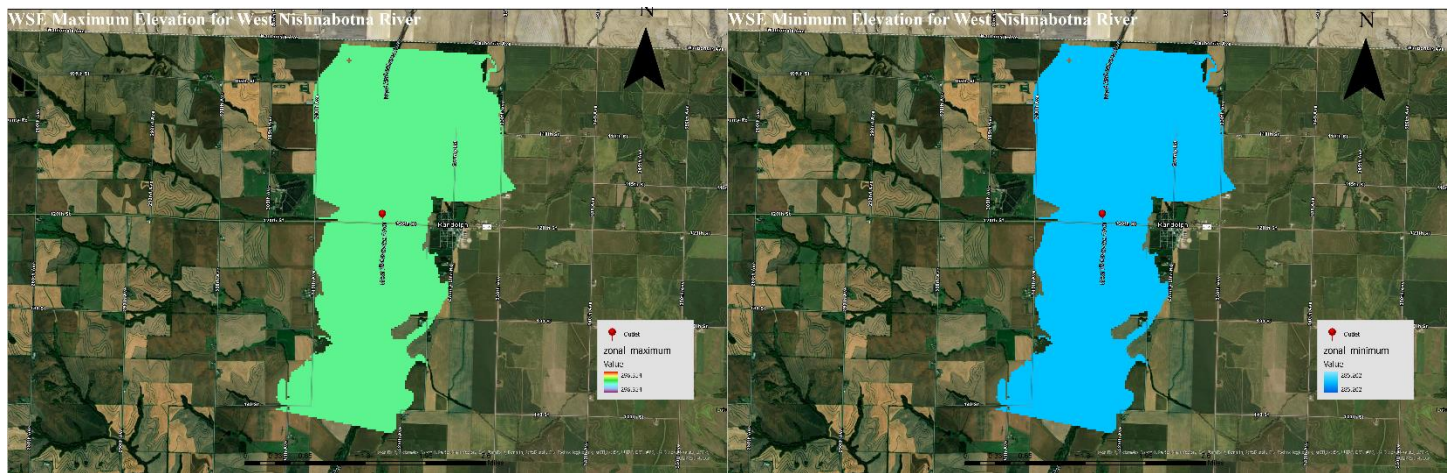


Figure 8 Maximum and Minimum Water Surface Elevation

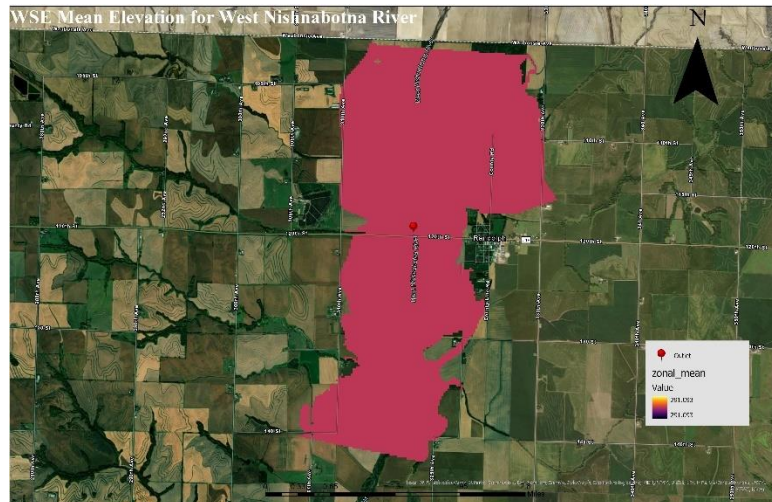


Figure 9 Mean Surface Water Elevation

The Water Surface Elevation (WSE) analysis for the West Nishnabotna River reveals critical insights into the flood dynamics. The **mean WSE of 291.093 meters** represents the average water level across the inundated area, providing an overall understanding of the floodplain's behavior. The **minimum WSE, measured at 285.202 meters**, highlights the lowest water levels within the flooded region, indicating areas with relatively shallow inundation and lower flood severity. On the other hand, the **maximum WSE of 296.354 meters** identifies the highest water levels reached during the flood event, corresponding to the most severely impacted areas.

## Conclusion:

This project successfully calibrated a hydrological model of the West Nishnabotna River Watershed, significantly improving the accuracy of simulated hydrographs. Manual calibration, which focused on refining parameters such as curve number, lag time, and Muskingum routing coefficients, proved more effective than automatic calibration, resulting in a 80% improvement in curve and lag times. The initial model achieved Percent Bias (PBIAS) of 1.94% and Nash-Sutcliffe Efficiency of 0.804 whereas with the new event validated model achieved excellent performance metrics, with a Percent Bias (PBIAS) of 0.03% and a Nash-Sutcliffe Efficiency (NSE) of 0.744, demonstrating its ability to closely replicate observed flow.

Through this project, valuable insights were gained into the hydrological behavior of the watershed, including its response to a 100-year storm event. The study emphasized the critical role of parameter adjustment in improving model accuracy and the importance of validation using historical data. Additionally, the project underscored the utility of integrated hydrological modeling for flood risk assessment and informed decision-making in watershed management. The findings from this study provide a robust framework for future applications in flood mitigation, land-use planning, and sustainable water resource management.

## References

- Cyberfaces. (2024). *Accessing Watershed Shapefile Using Python*. Retrieved from Cyberfaces: <https://www.cyberfaces.org/learn/module/263-accessing-watershed-shapefile/materials>
- Iowa Watershed Approach. (2024). *Iowa Watershed Approach*. Retrieved from Iowa Watershed Approach Watershed: <https://iowawatershedapproach.org/resources/ghost/west-nishnabotna-river/#:~:text=The%20river%20is%20used%20heavily,state%2Ddesignated%20water%20trail%20in>
- NOAA . (2024). *Hydrometeorological Design Studies Center* . Retrieved from NOAA Weather Service : [https://hdsc.nws.noaa.gov/pfds/pfds\\_map\\_cont.html?bkmrk=ia](https://hdsc.nws.noaa.gov/pfds/pfds_map_cont.html?bkmrk=ia)
- USGS. (2024). *USGS National Current Water Data for Nation*. Retrieved from USGS : <https://waterdata.usgs.gov/nwis/rt>
- USGS National Elevation. (2024). *NED* . Retrieved from NED : <https://apps.nationalmap.gov/downloader/>
- USGS National Land Cover . (2024). *Annual National Land Cover Database* . Retrieved from USGS National Land Cover : <https://www.usgs.gov/centers/eros/science/annual-national-land-cover-database>
- Weebly. (2024). *Wildlife & Plants - Flora and Fauna on the Water Trail*. Retrieved from Friends of the West Nishnabotna River : <https://friendsofthewestnish.weebly.com/wildlife--plants.html#:~:text=Though%20you%20likely%20won't,bison%20and%20elk%20viewing%20area.>

## Appendix:

| (Initial) The Rainfall Data |                    |
|-----------------------------|--------------------|
| Date and Time               | Precipitation (mm) |
| 2/23/2020 8:15              | 0.0395             |
| 2/23/2020 8:35              | 0.0595             |
| 2/23/2020 8:55              | 0.0895             |
| 2/23/2020 9:15              | 0.0915             |
| 2/23/2020 9:35              | 0.0895             |
| 2/23/2020 9:55              | 0.0865             |
| 2/23/2020 10:15             | 0.0875             |
| 2/23/2020 10:35             | 0.0805             |
| 2/23/2020 10:55             | 0.0795             |
| 2/23/2020 11:15             | 0.0655             |
| 2/23/2020 11:35             | 0.0595             |
| 2/23/2020 11:55             | 0.0495             |
| 2/23/2020 12:35             | 0.0395             |
| 2/23/2020 12:55             | 0.0395             |
| 2/23/2020 20:15             | 0.0595             |
| 2/23/2020 20:35             | 0.1095             |
| 2/23/2020 20:55             | 0.1395             |
| 2/23/2020 21:15             | 0.0395             |
| 2/23/2020 21:35             | 0.0395             |
| 2/23/2020 21:55             | 0.0595             |
| 2/23/2020 22:35             | 0.0895             |
| 2/23/2020 22:55             | 0.1095             |
| 2/23/2020 23:15             | 0.0595             |
| 2/23/2020 23:35             | 0.1095             |
| 2/23/2020 23:55             | 0.1895             |

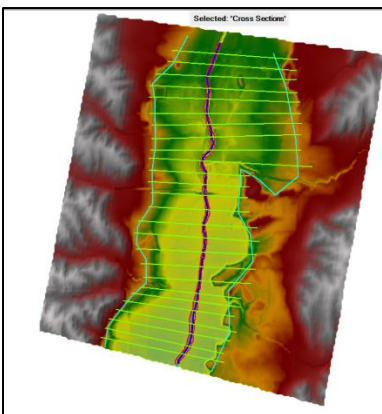


Figure 10 HEC-RAS digitization

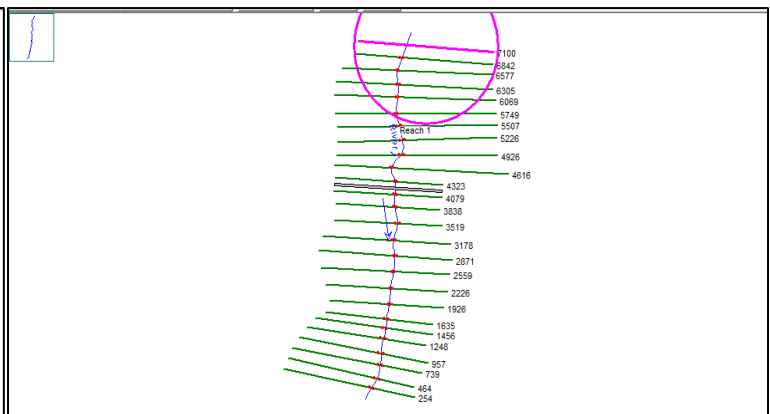


Figure 11 Bridge location between two 4323 and 3839